

TIC 224306120

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 4.175202 d
Epoch (t0) 1711.6742 BTJD
Duration 0.0840 d
Depth 1.231 % (12,313 ppm)
SNR 81.0

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 37 (gate ≥ 3)
Robust depth ratio 1.11 (gate ≥ 0.50)
Robust depth sigma 40.2 (gate ≥ 7.0)
Odd/even z-score 1.39 (consistent)
Secondary eclipse not detected

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 49 light-curve dip in this sector: 21.5 sigma t0 refined by 0.0050 phase
Difference-image peak SNR 5.54 (gate ≥ 5.0) Compactness 1.00 (gate ≥ 0.50)
Dimming centroid offset 0.65 px (14 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 224306117
5.01 px away, dTmag +3.78 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

